

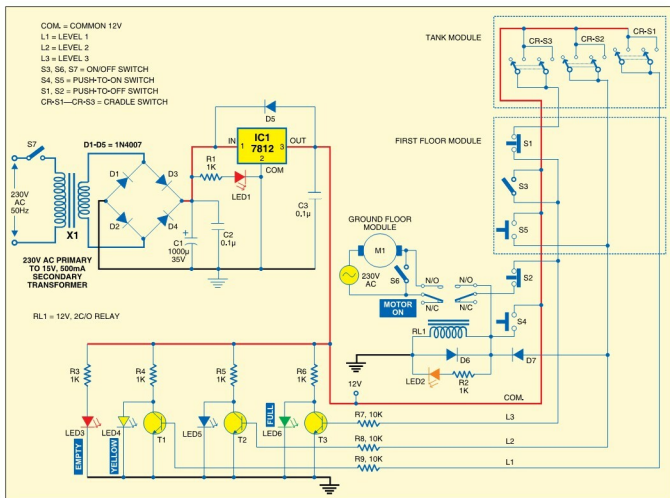


Fig. 2: Circuit of the automatic water pump controller

PARTS LIST

Semiconductors:

- IC1 - 7812, 12V regulator
- T1-T3 - BC547 npn transistor
- D1-D7 - 1N4007 rectifier diode
- LED1-LED6 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R6 - 1-kilo-ohm
- R7-R9 - 10-kilo-ohm

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2, C3 - 0.1 μ F ceramic disk

Miscellaneous:

- X1 - 230V AC primary to 15V, 500mA secondary transformer
- RL1 - 12V, 2C/O relay
- S1, S2 - Push-to-off switch
- S3, S6, S7 - On/off switch
- S4, S5 - Push-to-on switch
- CR-S1 - Telephone cradle switch
- CR-S3 - Telephone cradle switch

becomes high through cradle switch CR-S1 and 10-kilo-ohm resistor R9. Transistor T1 conducts and voltage across LED4 reduces. LED4 turns off. Similarly, transistors T2 and T3 conduct when their bases are high, turning off LED5 and LED6. Thus LED4 through LED6 turn off and only LED3 glows, indicating that the tank is empty. Note that LED3 plays the role of indicator for the presence of 12V DC as well as the water level. LED6 shows that the tank is full and LED4 and LED5 show the in-between levels.

The control unit consists of push-to-on switch S4, push-to-off switch S2, motor starter switch S6, diodes D6 and D7, and relay RL1. The connection is very simple as shown in Fig. 2. D6 acts as a free-wheeling diode. D7 is

used to prevent false glowing of level indicator LEDs during manual operation. LED2 indicates energisation of relay RL1.

First-floor module. The first-floor module consists of three switches and needs to be carefully connected to the two-core telephone cable. S1 is a push-to-off switch, S5 is a push-to-on switch and S3 is a simple on/off switch.

Tank module. The tank module consists of three telephone cradle switches (CR-S1 through CR-S3) and needs to be connected properly keeping in mind the number or colour of the wire.

Construction

Use plastic boards for the three switches of ground-floor and first-floor modules each (refer Fig. 1). In the